

EVALUATING THE POTENTIAL IMPACT OF WA DNR RECREATION CLOSURES

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Introduction

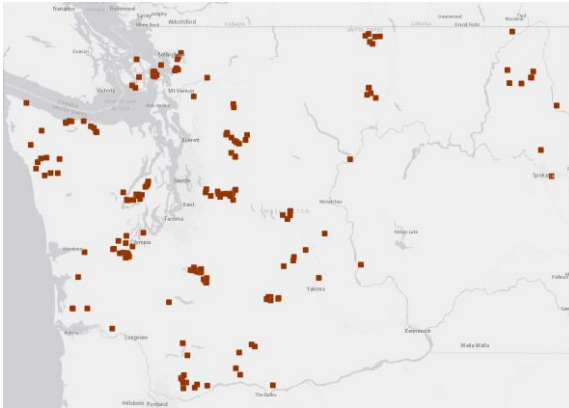
Project Sponsor Washington State Department of Natural Resources

Role: Manager for 200+ recreation sites, provide recreational activities, environmental conservation

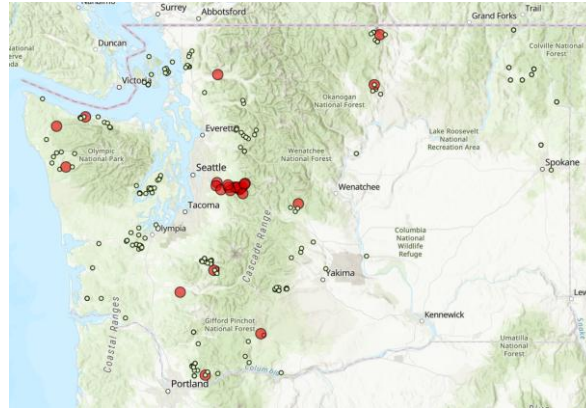
Background: Governor reduced funding for DNR, forcing closures and service reductions for over 25 sites

Need: Examine the potential impacts of site service reductions and closures

Project goal: To examine the spatial, environmental, and demographic impacts of site closures across WA



All recreation sites owned by DNR in WA State



Recreation sites color coded by closures



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Need: Examine the potential impacts of site impacts and closures

Project goal: To examine the spatial, environmental, and demographic impacts of site closures

This group intended to accomplish the following:

- Identify any disproportionate impacts on the basis of geography and demographics
- Identify alternative sites providing the same activities as nearby closed sites
- Analyze challenges to environmental equity through site closures and impacts
- Create datasets and layers for future use by DNR and other researchers

Our work is motivated by the following knowledge and information gaps:

- How have the spatial distribution of recreational sites changed before/after changes?
- How do different demographics experience unequal accessibility following changes?
- What activities have been impacted the most following closures/service reductions?
- What is the availability of alternative sites providing lost activities for impacted sites?



Project Methods & Approach

Software



Methods Proximity Analysis • Spatial Joins

Data Collection

- DNR sites
- USFS sites
- WA State Parks
- Census / ACS
- EHD V2 data

Data Prep & Cleaning

- Clean datasets
- Site classification
- Activity categorization

Spatial Analysis

- Alternative analysis
- Nearby accessibility modeling
- Equity and Health analysis

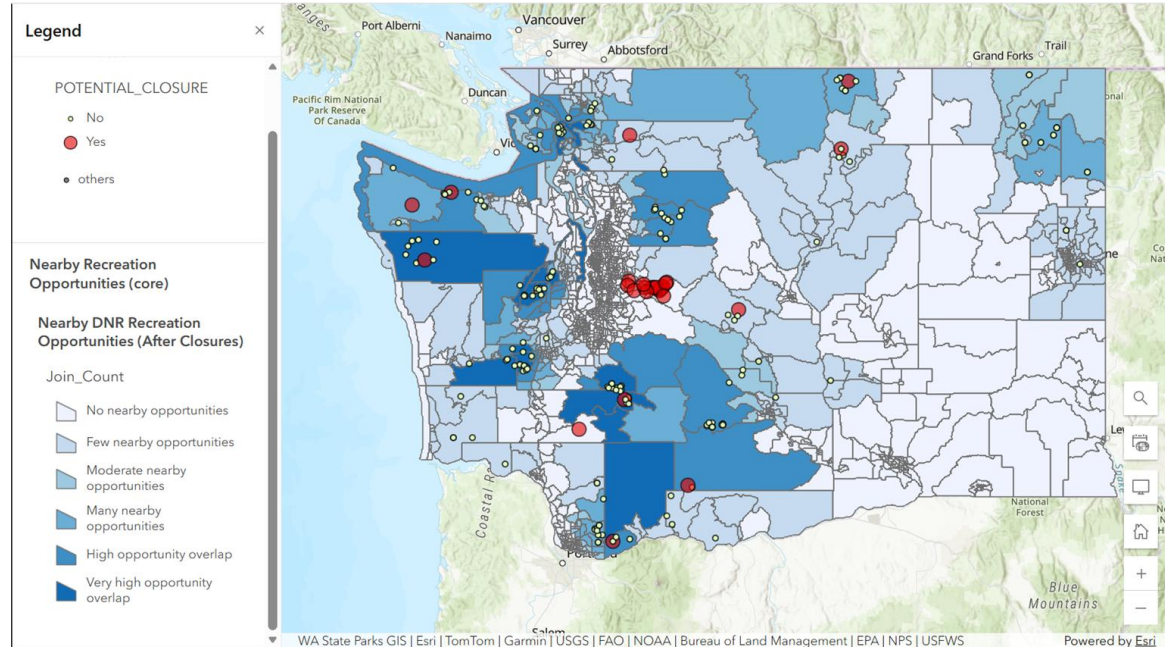
Findings

- DNR-only analysis
- Multiagency analysis
- Community impacts

Deliverables ArcGIS Online web map • Final layers • Final report • Data package for DNR

Project Results

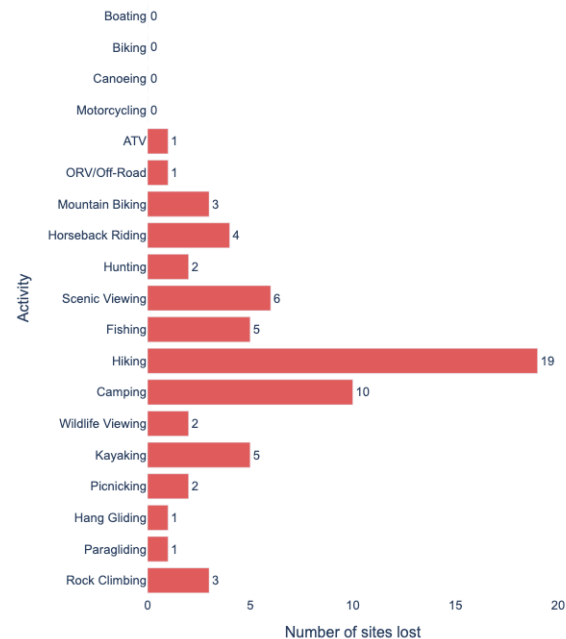
- Web Map
 - Web Layers for DNR Use
 - DNR Rec Closures
 - Layers identify alternative opportunities nearby closed sites
 - Evaluates the impact of closures by census tract
 - Alternative Site and Accessibility analysis



Conclusions and Recommendations

- Out of 11 full - closure sites, 3 have no relevant alternative locations
 - Bear Creek Campground (Clallam County)
 - Chopaka Lake Campground (Okanogan County)
 - Rock Lakes Campground (Okanogan County)
- 15 Snoqualmie locations will have seasonal toilet service reductions
 - No alternatives for activities
 - Inconvenience, but not severe quality reduction
- No significant change in activity disparity versus pre-closure
 - Access to hiking was most impacted
 - Still many alternatives
- Closure sites predominantly not in high-disparity regions
 - Potentially because access was limited to begin with
- **Recommendation:**
 - Prioritize the sites with no / few alternative locations
 - Our datasets include accessibility and socioeconomic disparity data
 - We hope these resources are useful for future considerations

Sites Lost Per Activity Using Final Closure List



Next steps

- Expand the number of activities investigated
 - We looked at top ~7 most affected activities
- Who uses the sites?
 - Some information was provided by sponsor, but currently unused
 - Would pair well with the information on environmental health considerations
- Transportation analysis
 - Accessibility by bus, car, walking
- Incorporate visitor numbers
 - Indicator of cost to community - should a site be shuttered?
- Maintenance costs
 - Indicator of cost to DNR - are the expenses justified?
 - Cost to visitor-count comparison
- Neighboring / bordering sites (BC, OR, ID)



QUESTIONS?